

POWER LINE NOISE DETECTION, ANALYSIS, AND LOCATION #2
REAGAN CARLTON, ALI HUSSEIN, ABIGAIL PURCHLA

CONCEPT OF OPERATIONS

REVISION 0
17 SEPTEMBER 2026

CONCEPT OF OPERATIONS FOR POWER LINE NOISE DETECTION, ANALYSIS, AND LOCATION #2

TEAM 35

APPROVED BY:

PROJECT LEADER _____ DATE _____

PROF. LUSHERDATE

T/A
DATE

Change Record

Table 1 List of changes

Rev.	Date	Originator	Approvals	Description
0	09/16/2026	Reagan Carlton, Ali Hussein, Abi- gail Purchla	-	First draft of CONOPS report.

Table of Contents

Table of Contents	III
List of Tables	IV
No table of figures entries found.....	IV
List of Figures	V
No table of figures entries found.....	Error! Bookmark not defined.
1. Executive Summary	1
2. Introduction	2
2.1. Background	2
2.2. Overview	3
2.3. Referenced Documents and Standards	4
3. Operating Concept.....	5
3.1. Scope	5
3.2. Operational Description and Constraints	5
3.3. System Description	6
3.4. Modes of Operations	7
3.5. Users	7
3.6. Support	8
4. Scenario(s)	9
4.1. Locating a Reported Power Line Noise Source	9
5. Analysis	9
5.1. Summary of Proposed Improvements	9
5.2. Disadvantages and Limitations	9
5.3. Alternatives	9
5.4. Impact	9

List of Tables

No table of figures entries found.

List of Figures

No table of figures entries found.

1. Executive Summary

The Power Line Noise Detection, Analysis, and Location capstone project focuses on developing and validating a portable, non-contact RF system for detecting power-line arcing and supporting source-location work. Power line sparks generate short-duration broadband electromagnetic emissions that can raise the local radio-noise floor, interfere with communications, and indicate defective or unsafe utility hardware. The end goal is a field-capable device that receives RF emissions with an antenna, converts them to digital I/Q data using a software-defined radio, analyzes the signal in Matlab/Simulink, and provides a directional indication to the user.

The project is divided into three independently validated subsystems. Student 1 is responsible for spark source and RF characterization, including source waveform modeling, timing, spectral behavior, antenna assumptions, and expected field strength. Student 2 owns RF acquisition and receiver hardware, including the SDR, antenna, laptop/GPS setup, metadata, and controlled bench verification. Student 3 is in charge of digital signal processing and detection, including I/Q processing, FFT/PSD analysis, spectrograms, feature extraction, detection metrics, and compute profiling.

During the fall semester, the team will validate these subsystems separately using simulation, recorded and generated data, and controlled bench testing. In the spring semester, the validated subsystems will be integrated into a complete field detection, analysis, and location prototype.

2. Introduction

The Power Line Noise Detection, Analysis, and Location project is focused on developing a portable, non-contact radio-frequency system for detecting electrical arcing or sparking on power-line equipment and supporting source location. Power-line arcing produces short-duration impulsive events with broadband RF content that can raise the local radio-noise floor, interfere with communications, and indicate damaged or unsafe utility hardware. The purpose of this CONOPS Report is to describe the intended operational concept, system scope, users, constraints, subsystem organization, and planned development path for the project.

The project is being developed using a subsystem-first approach. During ECEN 403, the team will focus on understanding the spark/noise signal, developing and validating the major subsystems, defining interfaces, and collecting evidence before committing to the final device architecture. Full system integration, field validation, and final prototype refinement are planned for ECEN 404.

2.1. Background

Power-line noise can be caused by arcing, sparking, loose hardware, damaged insulators, contaminated surfaces, or other defects on electrical distribution equipment. These events can create broadband electromagnetic emissions that propagate away from the source and may be detected remotely using an antenna and RF receiver. Because the proposed system is non-contact, it can support safer field investigation without requiring direct electrical connection to energized power-line equipment.

Prior research by Beasley provides the technical basis for modeling power-line spark noise. Beasley's work describes small, repetitive sparks as broadband RF noise sources with very fast current pulses and frequency components extending into the UHF and GHz range. The dissertation also shows that radiated spark noise can increase below 200 MHz, reach a broad maximum in approximately the 300–600 MHz region, and then decrease at higher frequencies. This supports the project's focus on RF-based detection and directional antenna evaluation.

Power-line environments are also challenging because they contain multiple forms of noise and interference. Power-line channels may include colored noise, impulsive noise, narrowband interference, background noise, fading, attenuation, multipath effects, and load-dependent variation. These effects make it necessary for the system to distinguish credible spark-related RF signatures from ordinary background or communication-channel noise.

Existing approaches for locating power-line noise rely on operator experience, handheld radios, or trial-and-error field searching. These approaches can be slow, imprecise, and dependent on the skill of the technician. The proposed system aims to improve this process by combining RF acquisition, signal processing, and directional/location support into an accessible, repeatable engineering tool.

2.2. Overview

The proposed system is designed to detect and analyze RF emissions associated with power-line arcing or sparking. The system consists of an antenna, software-defined radio (SDR), and MATLAB/Simulink-based signal-processing subsystem.

The antenna receives RF energy from the surrounding environment and provides the signal to the SDR. The SDR converts the received RF signal into digital in-phase and quadrature (I/Q) samples, which are then processed in MATLAB/Simulink. The processing subsystem performs time-domain and frequency-domain analysis, filtering, spectrogram generation, and feature extraction to identify characteristics associated with possible power-line sparking.

The processed results may include detection status, relative signal strength, event timing, and other measurements that can assist the user in determining the likely direction or location of the noise source. The three major project subsystems are described in the following sections.

The project is divided into three major subsystems:

2.2.1. Subsystem 1: Spark Source and RF Characterization

This subsystem defines the expected spark/noise source behavior. It includes research into pulse shape, relaxation behavior, timing relative to the power-system waveform, spectral content, RF radiation, antenna behavior, and expected field strength. Reagan owns this subsystem and is responsible for producing a documented source model and source-output interface.

2.2.2. Subsystem 2: Field Acquisition and Receiver Hardware

This subsystem receives RF signals and converts them into digital I/Q data for analysis. It includes the antenna, SDR receiver, laptop interface, receiver configuration, sampling parameters, metadata collection, and controlled bench verification. Abigail owns this subsystem and is responsible for producing a repeatable RF acquisition chain and a documented receiver-output interface.

2.2.3. Subsystem 3: Digital Signal Processing and Detection

This subsystem processes simulated or measured I/Q data. It includes signal conditioning, filtering, FFT/PSD analysis, spectrogram generation, feature extraction, detection logic, false-alarm testing, and compute profiling for later implementation decisions. Ali owns this subsystem and is responsible for converting I/Q data into reproducible detection results and performance metrics.

2.3. Referenced Documents and Standards

The following documents and references support the project definition, technical approach, and development plan:

2.3.1. Power Line Noise Detection, Analysis, and Location #2 — Team 2 Working Project Guide

This document defines the current mission, subsystem split, sponsor guidance, operating concept, validation framework, execution plan, and ECEN 403/404 development roadmap.

2.3.2. An Investigation of the Radiated Signals Produced by Small Sparks on Power Lines - Beasley Dissertation

This subsystem receives RF signals and converts them into digital I/Q data for analysis. It includes the antenna, SDR receiver, laptop interface, receiver configuration, sampling parameters, metadata collection, and controlled bench verification. Abigail owns this subsystem and is responsible for producing a repeatable RF acquisition chain and a documented receiver-output interface.

2.3.3. Noise and Interference in Power Line Channels – Penagos and Tache

This reference describes power-line channels as noisy and time-varying environments affected by attenuation, impedance variation, multipath, impulsive noise, narrow-band interference, background noise, and load-dependent behavior.

2.3.4. MATLAB/Simulink Signal Processing and Communications Toolbox Documentation

These resources support planned modeling, SDR acquisition, I/Q processing, signal analysis, QAM/AWGN benchmarking, and detection workflow.

2.3.5. Sponsor Guidance from Dr. Talley and Course Guidance from Dr. Lusher

Sponsor and instructor guidance define the current project direction: source modeling, SDR-based RF acquisition, MATLAB/Simulink analysis, controlled bench verification, GPS/time-location logging as an option, and full system integration during the second semester.

3. Operating Concept

3.1. Scope

The Power Line Noise Detection, Analysis, and Location #2 project will develop a method for detecting, analyzing, and eventually locating radio-frequency emissions produced by electrical arcing and sparking on power-line equipment. The system will use a combination of MATLAB/Simulink modeling, a software-defined radio (SDR), signal-processing algorithms, and a suitable antenna.

During ECEN 403, the main focus is to develop and verify the major subsystems independently. These include characterization of the spark/noise source, acquisition of RF signals using the antenna and SDR, and digital signal processing and detection using MATLAB/Simulink. The completed subsystems will later be integrated into a complete field detection system. Dr. Lusher clarified that the normal capstone approach is to build the subsystems during the first semester and integrate them into the complete system during the second semester.

The proposed system is intended to operate without physically contacting energized power-line equipment. RF emissions from sparking equipment will instead be detected remotely using an antenna and receiver.

3.2. Operational Description and Constraints

During operation, an antenna will receive electromagnetic emissions from the surrounding environment. The antenna output will be connected to an SDR receiver, which converts the received RF signal into digital in-phase and quadrature, or I/Q, samples. These samples will then be transferred to a laptop or other processing platform running the MATLAB/Simulink detection algorithm.

The processing subsystem will analyze the received signal in both the time and frequency domains. Techniques such as filtering, FFT or power spectral density analysis, spectrogram analysis, and transient feature extraction may be used to determine whether the received signal contains characteristics associated with power-line sparking. If a spark-related signal is detected, the system may provide information such as detection status, relative signal strength, event timing, or other measurements useful for locating the source.

The project currently considers an SDR capable of operation around 200 MHz with more than 100 kHz of receive bandwidth. Controlled testing may also use a signal generator or Pluto SDR to inject a known signal through coaxial cable and an attenuator directly into the receiver. This allows the receiver and detection algorithm to be tested before field measurements are performed.

Important operational constraints include intermittent spark activity, background RF interference, weak received signals, limited field data, SDR bandwidth and sampling limitations, and variation in the characteristics of different spark sources. The system must also be operated at a safe distance from energized electrical equipment.

3.3. System Description

The proposed system is divided into three major subsystems that can be developed and tested separately during ECEN 403.

The three subsystems are designed to operate as a connected signal chain. Subsystem 1 defines the expected spark and RF characteristics, Subsystem 2 acquires RF measurements and converts them into I/Q data, and Subsystem 3 processes those samples to produce detection results and performance metrics. After independent verification, the three subsystems will be integrated into the complete field detection and location system.

3.3.1. Subsystem 1: Spark Source and RF Characterization

This subsystem defines the expected electrical and electromagnetic characteristics of power-line sparking. Reference information from the Beasley dissertation, field measurements, and MATLAB/Simulink models will be used to understand the transient waveform, timing, frequency content, and expected radiated field strength.

3.3.2. Subsystem 2: RF Acquisition and Receiver

This subsystem receives the RF signal and converts it into digital data. The main components include the antenna, SDR receiver, laptop connection, and potentially GPS for recording the time and location of field measurements. The MATLAB Communications Toolbox will be used to communicate with the SDR and acquire I/Q samples. Dr. Talley specifically recommended that each group use an SDR with MATLAB/Simulink Signal Processing and Communications tools and suggested GPS capability for field measurements.

3.3.3. Subsystem 3: Digital Signal Processing and Detection

This subsystem processes the I/Q samples produced by the receiver. MATLAB and Simulink will be used to perform signal conditioning, filtering, time-domain analysis, frequency-domain analysis, spectrogram generation, feature extraction, and detection. The purpose of this subsystem is to distinguish possible sparking events from ordinary background or communication channel noise.

The project objective identifies two major development tracks: simulation and algorithm development, and development of a field device for gathering and recording signals. These tracks will later be integrated into the complete detection and location system.

3.4. Modes of Operations

The system will support several operating modes during development and testing.

3.4.1. Simulation Mode

MATLAB/Simulink will generate synthetic power-line and spark signals. This mode allows the team to test signal timing, processing methods, and detection algorithms without requiring live RF hardware.

3.4.2. Bench Verification Mode

A known waveform will be generated using laboratory equipment or a Pluto SDR and transferred through coaxial cable and an attenuator into the SDR. This provides a controlled end-to-end test of the acquisition and detection subsystems.

3.4.3. Recorded Data Analysis Mode

Previously recorded SDR I/Q data will be loaded into MATLAB and processed offline. This mode allows the team to develop and compare algorithms repeatedly using the same test data.

3.4.4. Field Acquisition and Detection Mode

The antenna and SDR will collect RF signals from actual field locations. The processing subsystem will analyze the received data and determine whether characteristics associated with power-line sparking are present.

Future versions of the system may also support a standalone field device using a Raspberry Pi or similar processing platform. The current project objective specifically includes evaluating both laptop-based and Raspberry Pi-based implementations.

3.5. Users

The expected users of the final system include utility engineers, electrical technicians, field maintenance personnel, researchers, and other personnel responsible for locating electrical interference or defective power-line equipment.

The final system should minimize the amount of specialized signal-processing knowledge required during normal field operation. A user should be able to connect the antenna and receiver, configure the required operating settings, begin collecting measurements, and view the detection results.

Engineering personnel responsible for setup, calibration, development, or troubleshooting will require additional knowledge of SDR operation, MATLAB/Simulink, RF measurements, and signal processing.

The project team will serve as the primary users during development and verification.

3.6. Support

Support for the system will be provided through project documentation, setup procedures, software documentation, test procedures, and troubleshooting information. These materials will be maintained throughout development so that each subsystem can be tested, reproduced, and later integrated into the complete system.

The team will maintain MATLAB scripts, Simulink models, SDR configuration information, test results, figures, and other project files in the shared project repository. The documentation will identify required software and hardware, subsystem connections, SDR settings, expected inputs and outputs, and the procedures required to repeat subsystem tests. Version control will be used so that validated versions of the source modeling, SDR acquisition, and signal-processing subsystems can be reproduced during integration and field testing.

A future user guide will describe the hardware setup and required connections, SDR and antenna configuration, MATLAB/Simulink startup procedures, and the steps required to collect and analyze data. It will also explain calibration and known-signal verification procedures, interpretation of detection results, and basic troubleshooting methods. This documentation is intended to allow another team member or future user to operate the system without needing detailed knowledge of how each subsystem was originally developed.

The support approach follows the overall project development process, which includes source modeling, SDR-based signal acquisition, signal analysis, detection algorithm development, controlled verification, subsystem testing, and later field-system integration. Maintaining clear documentation for each stage will help ensure that the individual subsystems can be independently verified before they are combined into the final system.

4. Scenario(s)

4.1. Locating a Reported Power Line Noise Source

A customer may report significant radio-frequency interference in an area where power-line noise is suspected. A technician would bring the portable detection system to the reported location, configure the antenna and SDR receiver, launch the detection software, and begin collecting measurements. The system would analyze the received RF data and provide information such as detection status and relative signal strength. The technician could then move or rotate the antenna while observing changes in the measured signal to determine the likely direction of the noise source. After the suspected source is identified and the appropriate utility personnel inspect or repair the equipment, the system could be used again to determine whether the detected interference has been reduced or eliminated.

5. Analysis

5.1. Summary of Proposed Improvements

The proposed system is intended to make the process of detecting and locating power-line noise more systematic and repeatable. Instead of relying only on operator experience or audible radio interference, the system will combine RF measurements with digital signal processing to identify signal characteristics associated with sparking. Directional measurements and signal-strength information may also help reduce the search area when locating a suspected source. Earlier detection of abnormal sparking may allow utility personnel to investigate equipment before the condition becomes more severe.

5.2. Disadvantages and Limitations

The system will have several practical limitations. Field operation may be limited by battery life, receiver sensitivity, antenna characteristics, environmental conditions, and background RF interference. Detection range will depend on the strength of the spark source, propagation conditions, receiver configuration, and surrounding noise. Multiple RF or spark sources in the same area may also make individual sources more difficult to distinguish. In addition, the system must be operated at a safe distance from energized power-line equipment, and field measurements may vary between locations and operating conditions.

5.3. Alternatives

One alternative is to install permanent monitoring sensors that continuously observe RF noise near power-line equipment. This approach could provide continuous measurements but would require additional hardware, installation, and maintenance. Another alternative is the traditional use of a handheld radio to search for the strongest audible interference. This method is inexpensive but depends heavily on operator experience and provides limited quantitative analysis. A drone-mounted receiver could also place the antenna closer to power-line equipment, but this approach would introduce additional hardware, control, safety, regulatory, and operational requirements.

5.4. Impact

The proposed system may help utilities identify abnormal sparking or arcing earlier and support inspection of potentially defective equipment. Earlier identification of these conditions

Power Line Noise Detection, Analysis, And Location #2

may reduce communication interference and provide additional information for maintenance decisions. The system may also reduce the amount of time required to search for an RF noise source by providing repeatable signal measurements. Potential concerns include interference from unrelated communication signals, limits in detection accuracy, and the need to operate the equipment safely near power-line infrastructure.